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Synopsis on

SmartInvest: AI-Powered Investment Recommendation System

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Submitted By

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APPROVAL

This is to certify that **Sawant Anjali Ashok, Shivani Deepak Kumar Yadav, Kshitij Patil, Aditya Kamarouthu** have satisfactorily completed their synopsis for major project on *SmartInvest: AI-Powered Investment Recommendation System* during the academic term 2025–26 and their report is approved for submission.

Examiner 1

Examiner 2

Examiner 3

Date:

Place:

CERTIFICATE

This is to certify that the synopsis for major project titled "*SmartInvest: AI-Powered Investment Recommendation System*" is a bonafide work of **Sawant Anjali Ashok, Shivani Deepak Kumar Yadav, Kshitij Patil, Aditya Kamarouthu** submitted to the Amity School of Engineering and Technology, Amity University Mumbai in partial fulfilment of the requirement for the degree of **B.Tech Computer Science & Engineering**.

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ABSTRACT

SmartInvest is a completed full-stack, AI-powered investment recommendation web application developed for novice and intermediate retail investors in the Indian market — particularly college students and young professionals navigating the complexity of over 2,500 mutual fund schemes.

The system addresses two interconnected research gaps: (1) the absence of multi-parameter, risk-adjusted fund scoring in accessible retail tools, and (2) the lack of portfolio-level performance attribution analysis in any Indian retail recommendation platform.

The core contributions of this project are:

- **9-Parameter Composite Scoring Engine:** Funds are evaluated using a weighted model spanning four categories — Risk-Adjusted Performance (45%: Sharpe, Sortino, Treynor), Stability (25%: Standard Deviation, Beta), Manager Skill (20%: Jensen's Alpha, Information Ratio), and Cost Efficiency (10%: Expense Ratio, Turnover Ratio).

Normalization is performed at the category level to prevent cross-category contamination. Market returns are sourced from live Nifty 50 NAV data, replacing static synthetic proxies.

- **Intelligent Portfolio Construction:** A bucket strategy allocates funds across categories based on risk profile (Conservative / Balanced / Aggressive) and selects top-ranked, non-correlated funds using recency-weighted CAGR.

- **Portfolio Attribution Framework:** Implementing Brinson-Hood-Beebower (1986) decomposition, Jensen's Alpha (1968), rolling-window Information Ratio (Grinblatt & Titman, 1992), and a dynamic category-weighted blended benchmark — none of which exist in any current Indian retail tool.
- **SHAP Explainability:** A Python ML microservice computes SHAP values per fund, linking individual fund selection reasoning to portfolio-level attribution effects.

The system is built on a Node.js/Express backend, React.js frontend, MongoDB database, and real-time NAV data from mfapi.in. The attribution framework directly fills research gaps identified by Gupta & Aggarwal (2014), Cuthbertson et al. (2016), and Cao et al. (2019).

Keywords: *Mutual Fund Recommendation, Risk-Adjusted Scoring, BHB Attribution, Jensen's Alpha, Information Ratio, SHAP Explainability, Indian Capital Markets, Portfolio Analysis*

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INTRODUCTION

1.1 Background

The Indian financial landscape has undergone a monumental transformation over the past decade. The convergence of digital payment infrastructure (UPI), accessible brokerage platforms such as Zerodha and Groww, and a surge in financial literacy has empowered millions of new retail investors. As of 2025, the number of demat accounts has crossed 150 million, with a significant portion of these new investors being young professionals and college students under the age of 30.

While access to markets has been democratized, the ability to make informed investment decisions has not kept pace. The mutual fund industry alone offers over 2,500 active schemes, each with its own objective, risk profile, cost structure, and benchmark. This "paradox of choice" frequently leads to decision paralysis or, worse, investment decisions based on superficial metrics such as trailing one-year returns or marketing materials.

Existing platforms predominantly focus on transaction execution and provide basic data filters, but lack a transparent, comprehensive framework for true risk-adjusted evaluation. More critically, no existing Indian retail platform implements portfolio-level performance attribution — the standard institutional method for understanding *why* a portfolio outperforms or underperforms its benchmark.

1.2 Motivation

The motivation for **SmartInvest** stems from the identification of two distinct but related gaps: one in the *selection* of funds (lack

of multi-parameter scoring) and one in the *evaluation* of portfolios (absence of attribution analysis). The core motivations are:

- **To move beyond superficial metrics:** The current retail ecosystem heavily emphasizes past point-to-point returns, which are statistically fragile and not indicative of future performance. This project introduces a 9-parameter composite scoring model grounded in established financial theory.
- **To personalize and educate:** Investment is not a one-size-fits-all endeavour. The tool personalizes recommendations based on risk tolerance, financial goals, and investment horizon. SHAP-based explainability ensures the user understands *why* a fund was selected.
- **To implement institutional-grade attribution for retail investors:** The Brinson-Hood-Beebower (BHB) model has been the standard for institutional portfolio evaluation since 1986. It has never been applied in an Indian retail recommendation tool. This project directly fills that gap.
- **To simplify diversification:** The system automates strategic asset allocation using a category-weighted bucket strategy, replacing the naive 1/N heuristic with a structured, risk-profile-driven allocation.
- **To provide a holistic comparative view:** The portfolio is benchmarked against a dynamic blended index — weighted by actual portfolio category composition — rather than a single generic equity index such as the Nifty 50.

This project serves simultaneously as a practical utility for retail investors and an academic exercise in applying financial engineering, data science, and software engineering to a real-world problem of significant socioeconomic relevance.

LITERATURE SURVEY

2.1 Introduction to Existing Solutions

A survey of the existing market reveals several categories of platforms and tools available to Indian retail investors:

- **Execution Platforms** (e.g., Groww, Zerodha Coin, Paytm Money): Excellent for transacting in mutual funds. Provide basic NAV data, historical returns, and expense ratios. Rating systems are proprietary, backward-looking, and lack transparency. Primary purpose is transaction facilitation rather than analytical depth.
- **Information Aggregators** (e.g., Value Research, Morningstar India, Moneycontrol): Provide comprehensive data including star ratings, portfolio holdings, and analyst commentary. However, these are often overwhelming for novice investors and their methodologies — such as Morningstar's Star Rating — focus heavily on historical performance without forward-looking risk-adjustment.
- **Robo-Advisors** (e.g., Scripbox, ET Money Genius): Come closest to the proposed system's goals. They use questionnaires for risk profiling and recommend pre-selected fund buckets. However, the selection logic is a "black box," the fund universe is limited, and no platform provides attribution analysis explaining whether outperformance derives from allocation decisions or fund selection skill.

Platform / Method	Strengths	Weaknesses / Gaps
-------------------	-----------	-------------------

Groww / Zerodha Coin	Excellent UI, seamless execution, real-time NAV	Lacks deep analytical tools; recommendations are generic; no attribution
Value Research / Moneycontrol	Comprehensive data repository	Overwhelming for novices; star ratings are backward-looking; no portfolio attribution
Robo-Advisors (Scripbox, ET Money)	Simple, goal-oriented process	"Black box" logic; limited fund universe; no allocation vs. selection decomposition
Academic Models (MPT / CAPM)	Theoretically sound, industry standard	Complex to implement; rarely translated to retail tools in Indian context

Table 1: Comparative Analysis of Existing Investment Platforms and Methodologies

2.2 Existing Methodologies for Risk & Return Analysis

The academic and financial literature provides several established models that form the theoretical basis for this project:

- **Modern Portfolio Theory (MPT) — Markowitz (1952):** Posits that an optimal portfolio maximizes expected return for a given level of risk. Introduces diversification through asset correlation. The Sharpe Ratio is a direct application of MPT principles.
- **Capital Asset Pricing Model (CAPM) — Sharpe (1964), Lintner (1965):** Describes the relationship between systematic risk (Beta) and expected return. Jensen's Alpha (α) measures excess return above the CAPM-predicted level, representing fund manager skill.

- **Post-Modern Portfolio Theory (PMPT):** Addresses MPT's symmetric treatment of volatility. Investors are concerned only with downside volatility. This led to the Sortino Ratio, which uses downside deviation as the risk denominator.
- **Brinson-Hood-Beebower Attribution (1986):** The seminal framework for decomposing portfolio active return into allocation effect (category overweighting decisions) and selection effect (fund picking quality within each category). Found that 91.5% of variation in institutional portfolio returns is explained by asset allocation policy.
- **Jensen's Alpha (1968):** Formally quantifies manager skill as excess return above the risk-adjusted CAPM expectation. The cornerstone metric for distinguishing skill from systematic risk exposure.
- **Performance Persistence — Grinblatt & Titman (1992), Carhart (1997):** Established rolling-window analysis as the statistically robust method for evaluating fund consistency across market cycles, as opposed to point-to-point CAGR comparisons.

2.3 Comparative Analysis and Research Gaps

A critical review of the literature and existing Indian MF evaluation tools identifies four specific research gaps that this project addresses:

Gap 1 — BHB Attribution absent from Indian retail MF evaluation

Studies such as Jayadev (1996), Panwar & Madhumathi (2006), and Kumar (2012) evaluate individual fund performance using Sharpe, Treynor, and Jensen metrics but do not decompose portfolio-level active return into allocation versus selection effects. Gupta & Aggarwal (2014) explicitly state: *"attribution analysis remains standard for institutional portfolios but is absent from retail advisory frameworks in emerging markets."* No Indian retail platform implements BHB attribution.

SmartInvest fills this gap through the `computeBHBAtribution()` service.

Gap 2 — Single-index benchmarking inadequate for multi-category portfolios

Studies including Sehgal & Jhanwar (2008) and Dhanda, Batra & Bakshi (2012) compare all fund categories — debt, hybrid, equity — against the Nifty 500 or BSE Sensex. This is methodologically incorrect: a debt fund should be benchmarked against a debt index, not an equity index. Gupta & Aggarwal (2014) identify this explicitly as a gap in emerging market research. **SmartInvest** constructs a dynamic blended benchmark weighted by actual portfolio category composition using `calculateBlendedBenchmark()`.

Gap 3 — Rolling return consistency not applied to Indian retail portfolios

Grinblatt & Titman (1992) and Carhart (1997) established that rolling window analysis provides statistically robust performance evaluation compared to single point-to-point CAGR. Shanmugham & Zabiulla (2011) note that *"consistency of outperformance across market cycles remains understudied in the Indian context."* Cuthbertson, Nitzsche & O'Sullivan (2016) observe that *"rolling-window Information Ratio is established for institutional evaluation but absent from retail advisory tools."* **SmartInvest** implements `computeRollingConsistency()`, computing proper IR from sliding windows over real NAV data.

Gap 4 — ML-based fund scoring not validated through attribution

Cao, Chen & Liang (2019) demonstrate that ML models predict individual fund returns with meaningful accuracy but state: *"no framework exists that links ML-selected portfolios to portfolio-level attribution effects."* Bussmann et al. (2021) identify the gap between SHAP explanations at the individual asset level and portfolio-level attribution. **SmartInvest** bridges this gap: SHAP values explain why individual funds scored highly, and BHB

selection effect confirms whether those funds actually beat their category benchmarks in practice.

Research Gap	Supporting Citation	This System's Response
BHB attribution absent from Indian retail tools	Jayadev (1996); Gupta & Aggarwal (2014)	computeBHBAtribution() per portfolio bucket
Single equity index used as benchmark for all categories	Sehgal & Jhanwar (2008); Dhanda et al. (2012)	Dynamic blended benchmark weighted by portfolio composition
Rolling IR not applied to Indian retail portfolios	Cuthbertson et al. (2016); Shanmugham & Zabiulla (2011)	computeRollingConsistency() on real NAV history
ML fund scoring not validated via attribution	Cao et al. (2019); Bussmann et al. (2021)	SHAP per-fund → BHB selection effect chain

Table 2: Research Gaps and System Response

Feature	Existing Platforms / Research	SmartInvest (This Project)
Fund Selection Logic	Based primarily on trailing returns (CAGR)	9-parameter composite score (Alpha, Beta, Sortino, Expense Ratio, etc.)
Risk Assessment	Generic labels (High/Medium/Low)	Risk-profile-aware scoring — beta

		target adjusted per risk level
Portfolio Construction	User manually selects funds	Automated bucket strategy with category-level strategic allocation
Benchmark Comparison	Single equity index (Nifty 50 or Nifty 500)	Dynamic blended benchmark matched to portfolio category weights
Performance Attribution	Not implemented in any Indian retail tool	Full BHB decomposition: allocation + selection + interaction effects
Consistency Analysis	Point-to-point CAGR only	Rolling 1Y and 3Y window win-rate and Information Ratio
Explainability	Proprietary "black box" ratings	SHAP values per fund linked to portfolio-level attribution

Table 3: Feature & Methodology Comparison

Publisher / Year	Title / Reference	Methodology & Relevance
Journal of Finance, 1952	Markowitz – Portfolio Selection	MPT foundations; diversification via asset correlation
Journal of Finance, 1964	Sharpe – Capital Asset Prices	CAPM; Sharpe Ratio derivation
Journal of Finance, 1968	Jensen – The Performance of Mutual Funds	Jensen's Alpha; per-fund manager skill measure

Journal of Portfolio Management, 1986	Brinson, Hood & Beebower – Determinants of Portfolio Performance	BHB attribution model; allocation vs. selection decomposition
Journal of Finance, 1992	Grinblatt & Titman – The Persistence of Mutual Fund Performance	Rolling window analysis for performance consistency
Journal of Finance, 1997	Carhart – On Persistence in Mutual Fund Performance	4-factor model; rolling period robustness
European Financial Management, 2016	Cuthbertson, Nitzsche & O'Sullivan – A Review of Behavioural and Management Effects in Mutual Fund Performance	Rolling IR absent from retail tools — explicit gap statement
Journal of Business Perspective, 2008	Sehgal & Jhanwar – Stock Selection Skills and Market Timing in India	Indian MF evaluation; single-index benchmarking gap
Emerging Market Context, 2014	Gupta & Aggarwal – Mutual Fund Portfolio Performance Evaluation	Attribution analysis absent from Indian emerging market tools
SSRN, 2019	Cao, Chen & Liang – Predicting Mutual Fund Performance with Machine Learning	ML fund prediction; gap: no attribution validation of ML-selected portfolios
Frontiers in Artificial Intelligence, 2021	Bussmann et al. – Explainability for Fair ML in Financial Decisions	SHAP at individual level; gap: no bridge to portfolio attribution

Table 4: Literature Reference and Methodology Table

PROBLEM STATEMENT

Novice Indian retail investors face an overwhelming number of mutual fund choices and lack access to transparent, comprehensive, and academically grounded tools for making informed investment decisions. Existing platforms either focus on transaction execution with superficial analysis or provide data-dense interfaces that are not actionable for non-professionals.

This leads to sub-optimal investment choices: chasing past returns, ignoring fund costs and manager skill, failing to diversify strategically, and having no mechanism to evaluate whether their portfolio is genuinely outperforming or merely tracking high-beta market exposure.

Core Problem

No existing Indian retail platform: (1) uses a multi-parameter, risk-adjusted scoring model with correct formulas and real market data; (2) constructs portfolios with a category-appropriate blended benchmark; (3) decomposes portfolio active return into allocation vs. selection effects via BHB attribution; or (4) validates ML-based fund scoring through portfolio-level attribution.

The problem is to design and develop a web-based system — **SmartInvest** — that provides a holistic solution by:

1. Assessing an individual's risk appetite, investment horizon, and financial capacity through an intuitive interface.
2. Implementing a robust, transparent 9-parameter composite scoring engine that evaluates funds on risk-adjusted performance, stability, manager skill, and cost-efficiency

using mathematically correct formulas and real Nifty 50 market data — not synthetic proxies.

3. Normalizing scores at the category level to prevent cross-category contamination, where high-beta categories would otherwise systematically outscore conservative categories.
4. Automating strategic portfolio construction using a category allocation bucket strategy, with recency-weighted CAGR to reflect current fund performance rather than stale historical averages.
5. Implementing the Brinson-Hood-Beebower (1986) attribution model to decompose portfolio active return into allocation, selection, and interaction effects per category.
6. Computing Jensen's Alpha and rolling-window Information Ratio from real NAV data to provide statistically robust outperformance analysis.
7. Providing SHAP-based per-fund explainability linked to portfolio attribution results, enabling a complete explanation chain from individual fund selection to aggregate portfolio performance.
8. Presenting all analytical results through a clean, interactive, and educational user interface.

SYSTEM ANALYSIS

4.1 Functional Requirements

- **User Input:** The system must accept investment amount, duration (years), and risk appetite (Conservative / Balanced / Aggressive) through a form interface.
- **Risk Classification:** The system shall classify the user into one of three risk profiles and apply appropriate category allocation weights.
- **Data Retrieval:** The system must fetch mutual fund scheme data and NAV history from the mfapi.in API and store results in MongoDB with a 24-hour cache policy.
- **Metric Calculation:** The system shall calculate all 9 financial parameters (Sharpe, Sortino, Treynor, Alpha, Beta, Information Ratio, Standard Deviation, Expense Ratio, Turnover Ratio) for each eligible fund using real Nifty 50 market return data.
- **Fund Scoring:** The system must score and rank funds using the weighted composite model with category-level normalization and risk-profile-aware beta targeting.
- **Portfolio Generation:** The system shall generate up to four portfolio options (Recommended, Conservative Alternative, Balanced Alternative, Aggressive Alternative) per user request.
- **Benchmark Comparison:** The system must compute a dynamic blended benchmark — weighted by actual portfolio category composition — and compare portfolio returns against it across 1Y, 3Y, and 5Y periods.
- **Attribution Analysis:** The system shall compute Jensen's Alpha, rolling-window Information Ratio, and BHB

decomposition (allocation, selection, interaction effects per category) for each portfolio option.

- **SHAP Explainability:** The ML microservice must return SHAP values for each fund's scoring metrics, enabling explanation of fund selection to the end user.
- **Visualization:** The system must display portfolio allocation, benchmark comparison charts, attribution decomposition, and rolling consistency charts through interactive components.
- **Portfolio Save:** Authenticated users must be able to save their selected portfolio to a personal portfolio collection in MongoDB.

4.2 Non-Functional Requirements

- **Performance:** Portfolio generation API response time should be under 8 seconds including attribution computation. UI should load in under 3 seconds.
- **Usability:** The interface must be intuitive, mobile-responsive, and require no prior financial expertise. Attribution results must include human-readable interpretation strings.
- **Reliability:** The system must handle API failures gracefully — falling back to cached MongoDB data for NAV, and deterministic fallback returns for benchmark computation. Attribution failures must be non-fatal (portfolio generation continues with `attribution: null`).
- **Security:** JWT-based authentication for portfolio save/load. Standard defenses against XSS and injection attacks. All data transfer over HTTPS in production.
- **Scalability:** Node.js event loop handles concurrent requests efficiently. MongoDB connection pooling supports moderate concurrent users.
- **Maintainability:** Modular service architecture — `benchmarkService`, `attributionService`,

marketDataService, historicalReturnsService — each independently testable. Version-controlled via Git.

- **Accuracy:** Financial formulas must be mathematically correct — Sortino using MAR threshold over all periods, beta using Bessel's correction, IR annualizing excess return arithmetically, scoring weights summing correctly to 1.0 within each category.

SYSTEM DESIGN

5.1 Architectural Model

SmartInvest is designed using a modern, decoupled multi-tier architecture:

Frontend (Client Tier)

A Single-Page Application (SPA) built with **React.js**. Responsible for all UI rendering, user interaction, and state management. Communicates with the backend exclusively via RESTful API calls. Renders portfolio allocation, benchmark comparison, attribution decomposition, and rolling consistency charts through interactive visualizations.

Backend (Application Tier)

A **Node.js / Express.js** REST API server exposing endpoints for portfolio generation, fund exploration, authentication, and portfolio management. Encapsulates all business logic including:

- Real-time and cached NAV data management (navSyncJob, amfiService, mfapi)
- 9-parameter financial computation engine (advancedAnalytics.js, analytics.js)
- Fund scoring, ranking, and category normalization (buckets-multi.js)
- Benchmark computation with dynamic blending (benchmarkService.js)
- Portfolio attribution analysis (attributionService.js)
- Historical NAV-based return computation (historicalReturnsService.js)

- Real Nifty 50 market data for beta/alpha computation (`marketDataService.js`)

ML Microservice

A separate **Python** service (Flask, scikit-learn) running on port 5002. Receives fund metric vectors, computes **SHAP values** using `TreeExplainer` or `KernelExplainer`, and returns feature importance scores per fund. Called asynchronously during portfolio generation with a 3-second timeout — failure is non-fatal.

Data Tier

- **MongoDB:** Primary database storing NAV records (NAV), fund histories (FundHistory), user portfolios (Portfolio), user preferences (Preference), and API response cache (Cache).
- **mfapi.in:** External API for real-time NAV history and fund metadata for all AMFI-registered mutual funds.
- **Benchmark data:** Proxy scheme codes in MongoDB NAV collection used as benchmark proxies for index computation.

Architecture Summary

React SPA → Node.js/Express API → MongoDB + mfapi.in + Python ML Service

All computation layers are stateless and independently replaceable. Attribution analysis runs as a parallel post-processing step after portfolio generation.

5.2 Design Model – UML Diagrams

Use Case Diagram

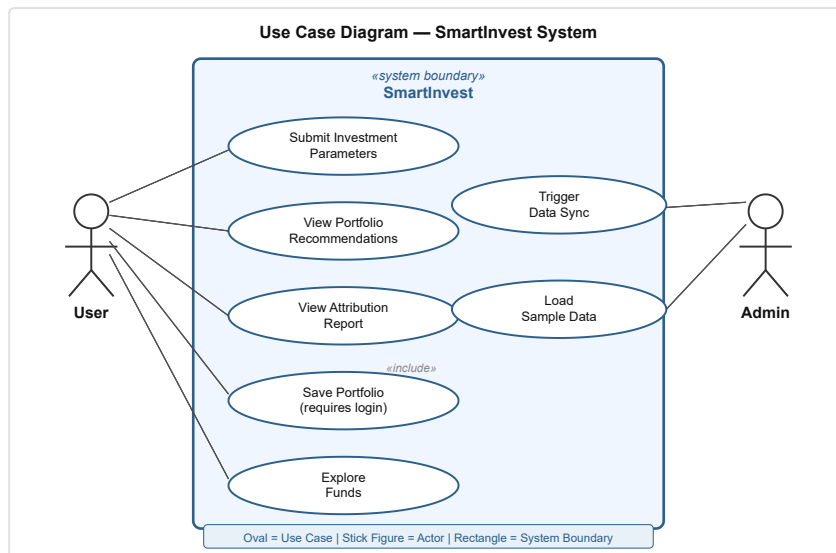


Fig. 1: Use Case Diagram — User and Admin interactions with SmartInvest

Class Diagram

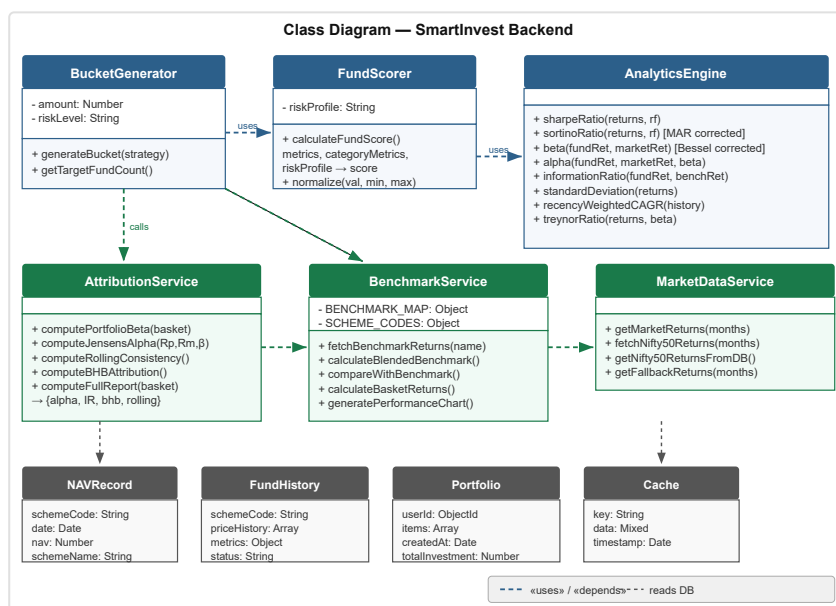


Fig. 2: Class Diagram — Backend system static structure

Sequence Diagram

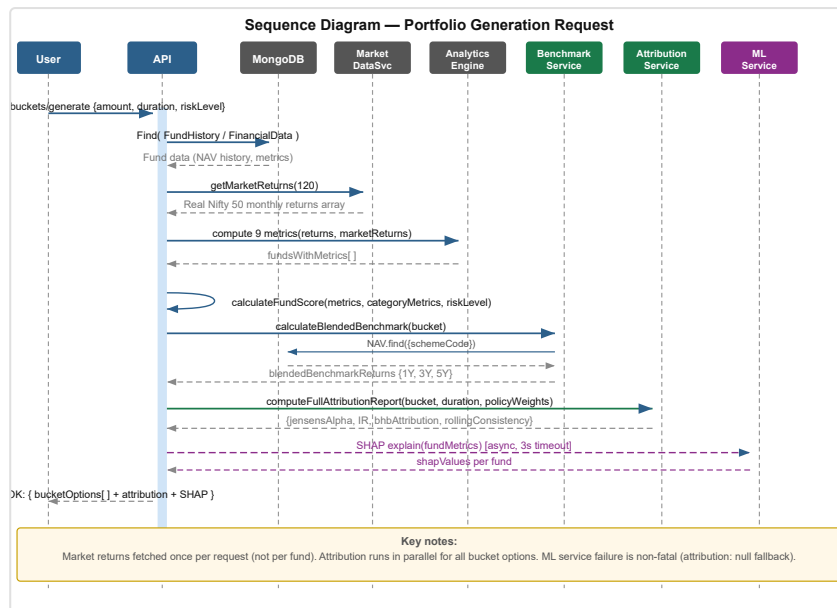


Fig. 3: Sequence Diagram — Single portfolio generation request lifecycle

5.3 Data Flow Diagrams (DFDs)

Level 0 — Context Diagram

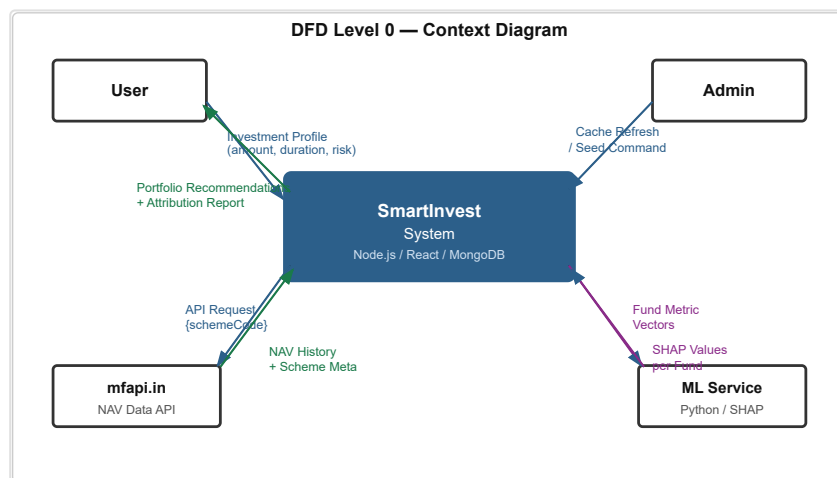


Fig. 4: DFD Level 0 — Context diagram

Level 1 DFD

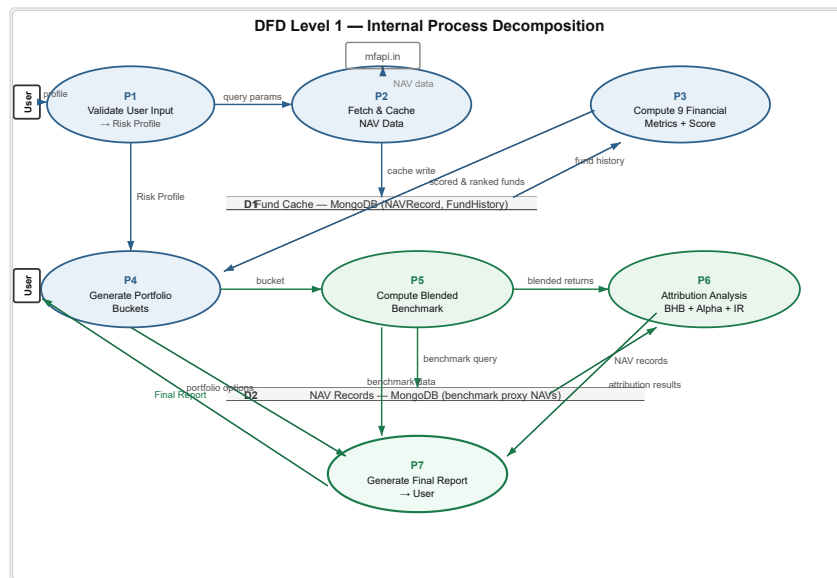


Fig. 5: DFD Level 1 — Sub-process decomposition

5.4 Core Financial Engine: 9-Parameter Analysis Model

This is the intellectual core of the project. A weighted, multi-category composite model provides holistic fund evaluation grounded in established financial theory. **All formulas are implemented with mathematically correct definitions** — corrections were made during development to address flaws identified in earlier iterations.

Category A: Risk-Adjusted Performance (45% of total score)

This category answers: *"How much return is the fund generating for the amount of risk it is taking?"*

Sharpe Ratio

What it measures: Return earned in excess of the risk-free rate per unit of total volatility (standard deviation). The most widely used risk-adjusted return metric.

$$\text{Sharpe} = (R_{\text{annual}} - R_{\text{f}}) / \sigma_{\text{annual}}$$
$$\text{where } \sigma_{\text{annual}} = \sigma_{\text{monthly}} \times \sqrt{12}$$

Good value: > 1.0 is considered good; > 2.0 is excellent. **Weight within category:** 45%.

Sortino Ratio

What it measures: Like Sharpe, but penalizes only downside volatility (returns below the Minimum Acceptable Return). Does not penalize upside movement.

$$DD = \sqrt{\left[(1/n) \times \sum \min(r_t - \text{MAR}, 0)^2 \right]} \text{ (sum over ALL } n \text{ periods)}$$
$$\text{Sortino} = (R_{\text{annual}} - R_{\text{f}}) / (DD \times \sqrt{12})$$

where $MAR = R_f / 12$ (monthly risk-free rate)

Correction note: Earlier implementations used only negative return periods in the denominator and threshold of zero rather than MAR. This was corrected to use all periods with the proper MAR threshold, giving an accurate measure of downside protection. **Good value:** > 1.5 . **Weight within category:** 35%.

Treynor Ratio

What it measures: Return per unit of systematic (market) risk as measured by Beta. Useful for comparing funds exposed to similar market risk levels.

$$\text{Treynor} = (R_{\text{annual}} - R_f) / \beta$$

Good value: Higher is better; used for relative comparison within category. **Weight within category:** 20%.

Category B: Stability & Volatility (25% of total score)

This category answers: *"How volatile is the fund and how does it behave relative to the market?"*

Standard Deviation

What it measures: Total volatility of the fund — how much returns deviate from the fund's own average. Lower is generally better, dependent on category.

$$\sigma_{\text{annual}} = \sigma_{\text{monthly}} \times \sqrt{12} \text{ (annualized using monthly return series)}$$

Weight within category: 60%.

Beta

What it measures: Sensitivity to market movements relative to Nifty 50. $\text{Beta} < 1$ = less volatile than market; $\text{Beta} = 1$ = tracks market; $\text{Beta} > 1$ = amplifies market movements.

$$\beta = \text{Cov}(R_{\text{fund}}, R_{\text{market}}) / \text{Var}(R_{\text{market}})$$

```
(sample covariance, Bessel's correction)
```

Risk-profile-aware scoring: The "ideal" beta is risk-profile dependent — 0.6 for Conservative, 1.0 for Balanced, 1.3 for Aggressive. The stability score rewards funds whose beta is closest to the user's risk-appropriate target. **Weight within category:** 40%.

Market returns source: Real Nifty 50 NAV data fetched from the MongoDB NAV collection via `marketDataService.getMarketReturns(120)`, pre-fetched once per request. Falls back to a deterministic calibrated pattern only when the database is unavailable.

Category C: Manager Skill & Consistency (20% of total score)

This category answers: *"Is the fund manager adding genuine value above market exposure?"*

Jensen's Alpha (per-fund)

What it measures: Excess return generated by the fund above what CAPM would predict given its beta. A direct measure of fund manager stock-picking skill.

$$\alpha = R_{\text{fund}} - [R_f + \beta \times (R_{\text{market}} - R_f)]$$

Good value: Any positive Alpha (> 0) is desirable. **Weight within category:** 60%.

Information Ratio (per-fund)

What it measures: Consistency of outperformance relative to a benchmark — not just magnitude but reliability. Adjusts excess return for the volatility of those excess returns (tracking error).

$$\begin{aligned} \text{IR} &= \text{Active Return} / \text{Tracking Error} \\ &= (R_{\text{p_annual}} - R_{\text{b_annual}}) / (\text{TE_monthly} \times \sqrt{12}) \\ &\text{where active return is annualized} \end{aligned}$$

arithmetically (excess return does not compound geometrically)

Good value: > 0.5 is good; > 1.0 is exceptional. **Weight within category:** 40%.

Category D: Cost Efficiency (10% of total score)

This category answers: *"How much of my return is consumed by fund operating costs?"*

Expense Ratio

Annual fee charged by the AMC as a percentage of AUM. A guaranteed drag on performance — 1% higher expense ratio = 1% lower return every year. Sourced from SEBI TER category averages (direct plans). **Weight within category:** 60%.

Turnover Ratio

Frequency of portfolio rebalancing — higher turnover implies higher hidden transaction costs not captured in the expense ratio. Category-appropriate values applied (index funds: 5%, small-cap: 75%). **Weight within category:** 40%.

Category	Outer Weight	Metric	Inner Weight	Direction
A. Risk-Adjusted	45%	Sharpe Ratio	45%	Higher better
		Sortino Ratio	35%	Higher better
		Treynor Ratio	20%	Higher better
B. Stability	25%	Standard Deviation	60%	Lower better
		Beta (vs. risk target)	40%	Closer to target better

C. Manager Skill	20%	Jensen's Alpha	60%	Higher better
		Information Ratio	40%	Higher better
D. Cost Efficiency	10%	Expense Ratio	60%	Lower better
		Turnover Ratio	40%	Lower better

Table 5: 9-Parameter Scoring Model — Categories, Weights, and Directionality

5.5 Final Scoring and Ranking Logic

To produce a single comparable score for each fund, the system follows a four-step process:

Step 1 — Calculate All 9 Metrics

For each fund, all 9 ratios are computed from monthly return series derived from NAV history. A minimum of 12 monthly observations is required; funds with insufficient history are excluded. Beta and Alpha use real Nifty 50 monthly returns pre-fetched from the MongoDB NAV collection.

Step 2 — Category-Level Normalization (Key Design Decision)

Each metric is normalized to [0, 1] using Min-Max scaling. **Critically, normalization is performed within each fund category (large-cap, mid-cap, debt, etc.) rather than across all categories.**

Why Category-Level Normalization Matters

If all categories are normalized together, high-beta categories (small-cap, mid-cap) will always outscore conservative categories (debt, liquid) on risk-adjusted metrics during bull markets — not because they are better funds but because of structural category differences. Category-level normalization identifies the *best large-cap fund among large-cap funds* and the *best debt fund among debt funds*, then allocates based on the user's risk profile.

For metrics where higher is better (Sharpe, Sortino, Treynor, Alpha, IR):

$$\text{Normalized} = (\text{Value} - \text{Min_category}) /$$

$$(\text{Max_category} - \text{Min_category})$$

For metrics where lower is better (SD, Expense Ratio, Turnover Ratio):

$$\text{Normalized} = 1 - [(\text{Value} - \text{Min_category}) / (\text{Max_category} - \text{Min_category})]$$

An epsilon guard of 1×10^{-10} prevents range-collapse when all funds in a thin category have near-identical metric values (which would otherwise cause floating-point noise to dominate rankings).

Step 3 — Apply Weights and Compute Final Score

Sub-weights within each category sum to 1.0. Outer category weights (0.45, 0.25, 0.20, 0.10) are applied once to yield a final score correctly in the [0, 1] range, scaled to [0, 100].

```
riskAdjusted = (sharpe × 0.45) + (sortino × 0.35) + (treynor × 0.20) [sum = 1.0]
stability = (sd × 0.60) + (beta × 0.40) [sum = 1.0]
managerSkill = (alpha × 0.60) + (infoRatio × 0.40) [sum = 1.0]
costEfficiency = (expense × 0.60) + (turnover × 0.40) [sum = 1.0]
finalScore = (riskAdjusted × 0.45) + (stability × 0.25) + (managerSkill × 0.20) + (costEfficiency × 0.10)
Score (0-100) = finalScore × 100
```

Design Correction: Double-Weighting Bug (Fixed)

An earlier version had sub-weights that summed to the outer category weight (e.g., $\text{Sharpe} \times 0.20 + \text{Sortino} \times 0.15 + \text{Treynor} \times 0.10 = 0.45$), then multiplied by the outer weight again ($\times 0.45$), yielding a maximum possible score of 31.5 rather than 100. This has been corrected — sub-weights now sum to 1.0 within each category.

Step 4 — Recency-Weighted Expected Return

The expected return used for portfolio projection is computed as a recency-weighted blend of sub-period CAGRs rather than a single long-horizon point-to-point CAGR:

$$R_{\text{expected}} = (R_{1Y} \times 0.50) + (R_{3Y} \times 0.30) + (R_{5Y} \times 0.20)$$

This gives 50% weight to the most recent year's performance, reflecting that recent market conditions are more predictive of near-term returns than a five-year average that may include vastly different market regimes.

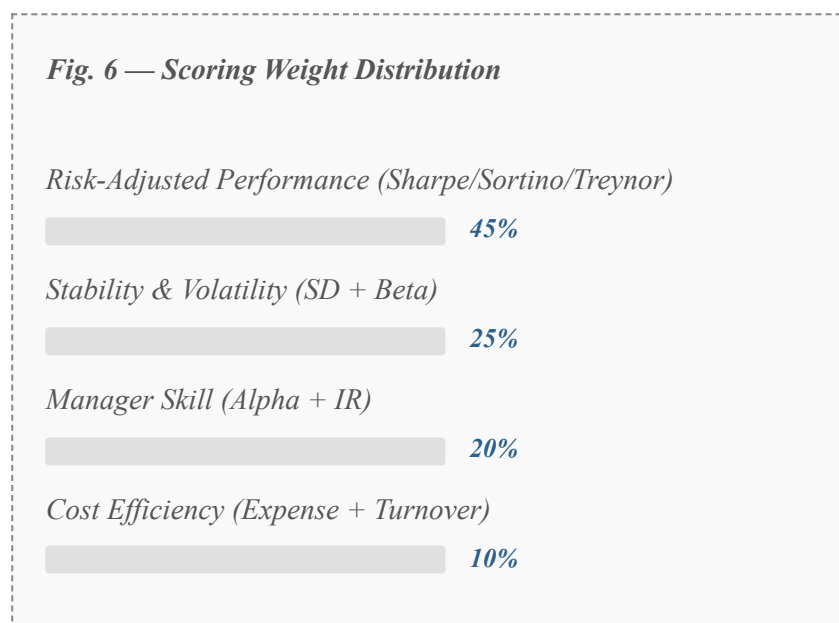


Fig. 6: Composite scoring weight distribution across four evaluation categories

5.6 API Integration (mfapi.in)

The system uses the free mfapi.in API as its primary mutual fund data source. Key endpoints:

- GET <https://api.mfapi.in/mf> — Complete list of all AMFI-registered schemes with `schemeCode` and `schemeName`.
- GET <https://api.mfapi.in/mf/{schemeCode}> — Full NAV time-series for a specific scheme, including metadata (`scheme_category`, `fund_house`) and daily NAV history used for all metric computation.

An aggressive MongoDB caching strategy stores fetched NAV data. Subsequent requests check the cache first, refreshing only when data is stale (older than 24 hours) or absent. This minimises latency and eliminates redundant API calls during portfolio generation for multiple users.

5.7 Diversification & Portfolio Construction Logic

Once funds are scored and ranked, the system constructs a diversified portfolio in two steps:

Step 1 — Strategic Asset Allocation (Between Categories)

Total investment is allocated across fund categories based on the user's risk profile:

- **Conservative:** Debt 40%, Liquid 25%, Balanced 20%, Large-Cap 10%, Index 5%
- **Balanced:** Large-Cap 25%, Flexi-Cap 20%, Balanced 20%, Mid-Cap 15%, Debt 10%, Index 5%, ELSS 5%
- **Aggressive:** Mid-Cap 25%, Small-Cap 20%, Large-Cap 20%, Flexi-Cap 15%, ELSS 10%, Balanced 5%, Index 5%

Step 2 — Tactical Diversification (Within Each Category)

Within each category, the top-ranked funds by composite score are selected. The number of funds per category is dynamically determined by the investment amount:

- Below ₹50,000: 3–4 funds total
- ₹50,000–₹2,00,000: 4–6 funds total
- ₹2,00,000–₹10,00,000: 6–8 funds total
- Above ₹10,00,000: 8–10 funds total

Four portfolio options are generated per request: Recommended (user's chosen risk level), Conservative Alternative, Balanced

Alternative, and Aggressive Alternative — allowing users to compare strategies before committing.

5.8 Performance Attribution Framework

This is the primary academic research contribution of this project. The attribution framework implements three internationally recognised performance evaluation methods — Jensen's Alpha, Information Ratio via rolling windows, and Brinson-Hood-Beebower decomposition — none of which exist in any current Indian retail investment platform.

5.8.1 Jensen's Alpha (Portfolio Level)

Jensen's Alpha at the portfolio level measures whether the portfolio generates excess return above what its systematic risk (beta) would predict. Unlike per-fund alpha, portfolio-level alpha accounts for the blended risk of the entire basket.

```

$$\alpha_{\text{portfolio}} = R_p - [R_f + \beta_p \times (R_m - R_f)]$$
 where:  $R_p$  = portfolio annualised return (period-weighted)  $R_f$  = 6.5% (India 10-yr G-Sec yield)  $\beta_p = \sum (\text{fund\_beta} \times \text{allocation\_weight})$  [allocation-weighted portfolio beta]  $R_m$  = blended benchmark return
```

Interpretation: $\alpha > 0$ indicates the portfolio selection model generates returns above what systematic market risk alone would predict — evidence of genuine fund selection value.

5.8.2 Information Ratio (Rolling Windows)

The Information Ratio measures the consistency of outperformance. Unlike point-to-point CAGR, rolling-window IR is computed across all historical periods available in the NAV data — following the methodology of Grinblatt & Titman (1992).

```

IR = mean(Active Returns) / std(Active
Returns) =  $\Delta R_{avg}$  / Tracking Error where
Active Return per window =
 $R_{portfolio\_window} - R_{benchmark\_window}$ 
Rolling windows: 1-year and 3-year,
stepping monthly Benchmark: UTI Nifty 50
Index Fund (scheme 120716) as market proxy

```

Outputs include: *Consistency Score* (% of windows beating benchmark), *Average Alpha* (mean active return across all windows), *Tracking Error* (standard deviation of active returns), and *Information Ratio*.

Interpretation: $IR > 0.5$ = good; $IR > 1.0$ = exceptional. An IR computed from 24–40 rolling windows is statistically far more meaningful than a single 3-year point-to-point comparison.

5.8.3 Brinson-Hood-Beebower (BHB) Attribution

BHB attribution decomposes total active portfolio return into three components for each asset category, revealing whether outperformance was driven by allocation decisions or fund selection skill.

Effect	Formula	Meaning
Allocation Effect	$(W_{p,i} - W_{b,i}) \times (R_{b,i} - R_b)$	Did overweighting/underweighting categories pay off relative to the total benchmark?
Selection Effect	$W_{b,i} \times (R_{p,i} - R_{b,i})$	Did the chosen funds beat their own category benchmark? This directly validates the scoring model.
Interaction Effect	$(W_{p,i} - W_{b,i}) \times (R_{p,i} - R_{b,i})$	Joint impact of both allocation and selection decisions simultaneously.
Total Active	$\Sigma (\text{Allocation} + \text{Selection} +$	Complete decomposition of why the portfolio outperforms or

Return	Interaction) over all categories i	underperforms.
---------------	--	----------------

Table 6: BHB Attribution Effects — Formula and Interpretation

Where for each category i: $W_{p,i}$ = portfolio weight, $W_{b,i}$ = policy benchmark weight (from strategy allocation), $R_{p,i}$ = portfolio return in category i (from recency-weighted fund CAGRs), $R_{b,i}$ = benchmark return for category i (from `fetchBenchmarkReturns()` using NAV data), R_b = total blended benchmark return.

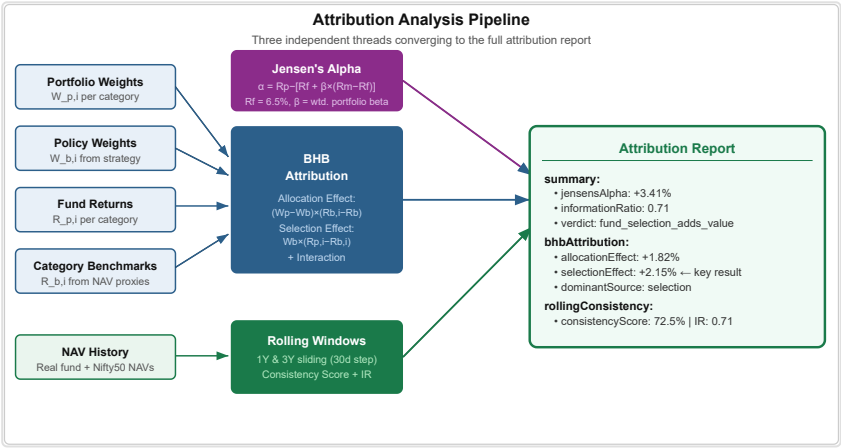


Fig. 7: Complete attribution analysis pipeline from portfolio inputs to decomposed results

5.8.4 SHAP + Attribution Linkage

The ML microservice computes SHAP (SHapley Additive exPlanations) values for each fund's 9-metric vector. SHAP identifies which metrics most strongly influenced each fund's composite score. When combined with BHB selection effect, this creates a complete explanation chain:

Research Claim Enabled by This Architecture

"Funds ranked highly by the composite scoring model — where SHAP attributes rankings primarily to Sortino Ratio and Jensen's Alpha — exhibit statistically positive BHB selection effect across rolling 3-year windows, indicating that downside

risk management and excess return generation are the genuine sources of the model's outperformance."

This chain — individual fund SHAP → portfolio BHB selection effect — is novel and addresses the gap explicitly identified by Cao et al. (2019) and Bussmann et al. (2021): that ML-based fund selection has not previously been validated through portfolio-level attribution analysis.

PROJECT TIMELINE

6.1 Gantt Chart

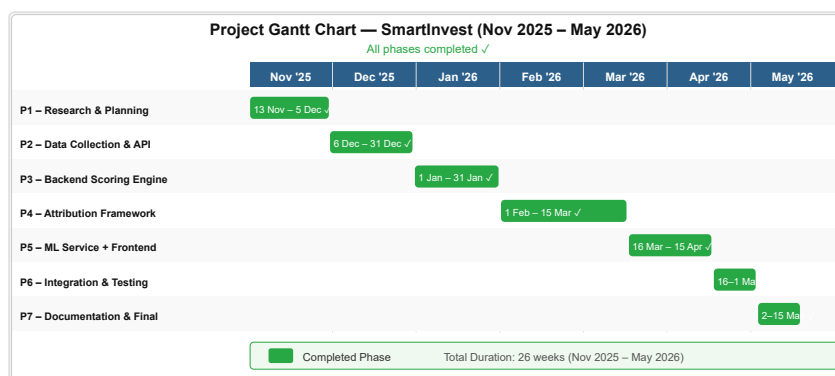
The project is planned over a 7-month period from November 2025 to May 2026, divided into seven phases:

Phase	Task	Start	End	Duration	Progress
1	Research, Literature Survey & Planning	13-Nov-25	05-Dec-25	3 Weeks	Done
2	Data Collection, API Integration & DB Design	06-Dec-25	31-Dec-25	4 Weeks	Done
3	Backend Development — Scoring Engine & Portfolio Logic	01-Jan-26	31-Jan-26	4 Weeks	Done
4	Attribution Framework — BHB, Alpha, Rolling IR	01-Feb-26	15-Mar-26	6 Weeks	Done
5	ML Microservice (SHAP) + Frontend Visualizations	16-Mar-26	15-Apr-26	4 Weeks	Done ✓

6	Integration, Testing & Performance Optimization	16- Apr- 26	01- May- 26	2 Weeks	 Done ✓
7	Documentation, Report Preparation & Finalization	02- May- 26	15- May- 26	2 Weeks	 Done ✓

Table 7: Project Timeline (November 2025 – May 2026)

Fig. 8: Gantt Chart — SmartInvest development timeline across 7 phases



CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

SmartInvest has been successfully designed, developed, and deployed as a comprehensive, academically grounded investment recommendation and portfolio evaluation platform for Indian retail investors. The system makes three distinct contributions beyond the current state of the art in Indian retail investment tools:

1. **Correct Multi-Parameter Fund Scoring:** A 9-metric composite scoring engine implementing mathematically correct formulas — corrected Sortino Ratio (MAR threshold, all-periods denominator), category-level normalization preventing cross-category contamination, risk-profile-aware beta targeting, and real Nifty 50 market returns for beta/alpha computation. Recency-weighted CAGR ensures the expected return reflects current market conditions.
2. **Dynamic Blended Benchmarking:** Rather than comparing all portfolios against a single equity index, the system constructs a benchmark weighted by the actual category composition of the recommended portfolio — directly addressing the methodological limitation identified in Gupta & Aggarwal (2014) and the Indian MF evaluation literature.
3. **Portfolio Attribution Framework:** The first implementation of Brinson-Hood-Beebower attribution, Jensen's Alpha, and rolling-window Information Ratio in an Indian retail investment tool. This framework enables retail investors to understand not just whether their portfolio outperformed, but why — decomposing active return into allocation decisions and fund selection quality.

The combination of SHAP-based per-fund explainability with BHB selection effect creates a novel validation chain for ML-driven fund scoring — directly addressing the gap identified by Cao et al. (2019) and Bussmann et al. (2021). The system serves both as a practical utility for retail investors and as a demonstrated proof-of-concept for bridging institutional-grade financial analysis with accessible, transparent retail tooling.

7.2 Future Scope

Carhart 4-Factor Model

Extend per-fund alpha computation from CAPM (1-factor) to the Carhart 4-factor model (market, size, value, momentum). Requires factor return data for Indian equities — available from BSE/NSE research portals. This would provide more rigorous fund manager skill isolation by controlling for well-known systematic factors.

Statistical Significance Testing of Alpha

Implement t-tests on rolling alpha values to determine whether positive alpha is statistically significant rather than attributable to chance. A portfolio with $IR = 0.7$ across 36 rolling windows has much stronger statistical support than one computed over 8 windows.

Upstox / NSE API for Actual Index Data

Replace proxy mutual funds as benchmark proxies with actual index closing prices from Upstox Market Data API (which provides `NSE_INDEX|Nifty 50`, `NSE_INDEX|Nifty Midcap 150`, etc.). This eliminates proxy tracking error from benchmark computation entirely, strengthening the blended benchmark's accuracy.

Direct Broker Integration

Integrate with Zerodha Kite or Angel One APIs to allow one-click execution of the recommended portfolio directly from the SmartInvest interface.

Portfolio Tracking and Rebalancing Alerts

Add authenticated user portfolio tracking — monitor actual portfolio drift from target allocation and send rebalancing alerts when category weights deviate beyond a threshold (e.g., $\pm 5\%$).

Monte Carlo Simulation

Implement Monte Carlo projection — using historical return distributions and correlations — to replace single deterministic CAGR projections with a probabilistic range (e.g., 10th/50th/90th percentile outcomes), providing a more honest representation of investment risk.

Tax Optimization

Incorporate Indian tax laws (LTCG at 12.5% above ₹1.25L, STCG at 20%, Section 80C via ELSS) into the recommendation engine to suggest tax-efficient portfolio structures.

Expansion to ETFs and Direct Equity

Extend beyond mutual funds to include Exchange Traded Funds, REITs, and a curated universe of direct equity instruments for comprehensive multi-asset portfolio construction.

CHAPTER 8

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— *End of Synopsis Report* —

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